

# Quantitative Amyloid Brain Imaging: A Literature Review of the Centiloid Scale in Alzheimer Disease Evaluation

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Amyloid PET has become a pivotal imaging biomarker for Alzheimer disease (AD), enabling in vivo detection and quantification of  $\beta$ -amyloid deposition. However, variability in quantitative measurements across tracers, scanners, and analysis methods has considerably limited direct comparison of amyloid burden between studies and centers. To address this issue, the Centiloid Project has been developed to provide a standardized quantitative scale for amyloid PET, harmonizing results across tracers and institutions. In this literature review, we aim to summarize current evidence on the development, validation, and clinical application of the Centiloid scale in amyloid PET imaging, emphasizing its methodologic foundations, tracer-specific conversions, and diagnostic thresholds. **Methods:** A literature review was conducted using PubMed, Scopus, and Embase databases from January 2015 through October 2025. Search terms included “amyloid PET,” “Centiloid,” “SUVR,” and “Alzheimer disease.” We reviewed articles with quantitative amyloid PET analyses and Centiloid conversion and studies which compared multiple tracers using MIMneuro software Centiloid calibration. **Results:** The Centiloid method demonstrates strong intertracer and interscanner harmonization when standardized to the  $^{11}\text{C}$ -Pittsburgh compound B reference, with high correlation ( $r^2 > 0.9$ ) across  $^{18}\text{F}$ -labeled tracers, including florbetapir, flutemetamol, and florbetaben. Most articles identified a Centiloid threshold between 20 and 25 Centiloids as indicative of significant amyloid pathology. Implementation of the Centiloid framework improved comparability across clinical trials and longitudinal studies, facilitating integration into the AT(N) research framework. **Conclusion:** The Centiloid scale represents a critical advancement in quantitative amyloid PET imaging, providing a universal reference that enhances data reproducibility, cross-trial comparisons, and clinical decision-making. Continued work is needed to expand standardization for novel tracers and hybrid imaging systems, ensuring full clinical translation of this metric in dementia imaging.

**Key Words:** amyloid PET; Centiloid; Alzheimer disease; SUVR

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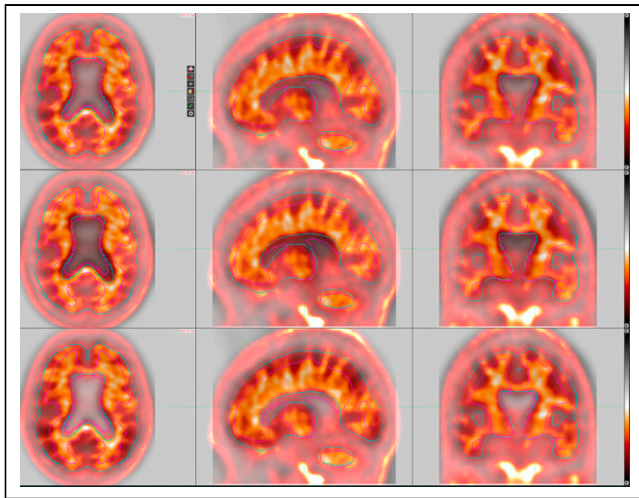
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Alzheimer disease (AD) is the most common cause of dementia, characterized neuropathologically by extracellular  $\beta$ -amyloid ( $\text{A}\beta$ ) plaques and intracellular neurofibrillary tangles. For decades, the definitive diagnosis of AD relied on postmortem histopathologic confirmation. The development of PET tracers that bind fibrillar  $\text{A}\beta$  has revolutionized clinical and research assessment by allowing direct visualization of amyloid deposition in vivo. Amyloid PET imaging now plays a pivotal role in patient stratification, early disease detection, and the evaluation of anti-amyloid therapies.

Quantification of amyloid burden using PET typically involves calculating the SUV ratio (SUVR), which compares tracer uptake in cortical target regions to that in a reference region devoid of amyloid. However, SUVRs are influenced by multiple factors, including scanner type, image reconstruction parameters, tracer kinetics, and selection of reference regions of interest (ROI). Furthermore, the introduction of various  $^{18}\text{F}$ -labeled tracers such as florbetapir, flutemetamol, and florbetaben, while broadening clinical accessibility, has introduced additional variability, as each tracer exhibits unique binding characteristics and dynamic properties. As a result, direct comparison of SUVRs between tracers and across sites has been unreliable, limiting multicenter and longitudinal study interpretation. Figure 1 demonstrates the ROIs for the cerebral hemispheres for white and gray matter and cerebellum as the reference using MIMneuro (MIM Software Inc.).

To overcome these challenges, the Centiloid Project, initiated by Klunk et al. in 2015, proposed a universal quantitative framework for amyloid PET standardization (1). The Centiloid (CL) scale defines 0 CL as the average amyloid burden in young, amyloid-negative controls and 100 CL as the mean value in patients with typical mild AD imaged with  $^{11}\text{C}$ -Pittsburgh compound B ( $^{11}\text{C}$ -PiB). This linear scaling allows SUVRs from any amyloid tracer to be converted to a common Centiloid unit using validated transformation equations derived from paired tracer studies. Consequently, the Centiloid framework enables harmonized quantification across tracers, scanners, and analysis pipelines, improving reproducibility and facilitating the aggregation of

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**FIGURE 1.** ROIs of cerebral hemispheres for white and gray matter and cerebellum as reference using MIMneuro.

large datasets. Depending on the amyloid burden in the brain, values less than 0 CL and more than 100 CL are possible. Figure 2 demonstrates a sample report generated by MIMneuro including regional ROIs, SUVRs, and Centiloid score.

Since its introduction, the Centiloid scale has been increasingly adopted in both research and clinical settings. Numerous studies have validated Centiloid conversions for the major  $^{18}\text{F}$ -labeled tracers, demonstrating high correlation ( $r^2 > 0.9$ ) with the  $^{11}\text{C}$ -PiB reference standard. The Centiloid scale also provides a consistent threshold for amyloid positivity, typically in the range of 20–25 CL, corresponding to the pathologic onset of significant A $\beta$  accumulation. Importantly, the use of Centiloid values has enabled direct comparison of amyloid burden

across clinical trials, enhanced interpretation of longitudinal changes, and facilitated integration with complementary biomarkers within the AT(N) framework (amyloid, tau, neurodegeneration) (2–5).

Despite these advances, challenges remain. Variability persists due to differences in preprocessing pipelines, ROI definitions, and scanner calibration procedures. Moreover, as newer PET tracers and hybrid PET/MR systems emerge, further standardization and validation will be necessary to maintain consistency with the Centiloid framework. Nonetheless, the Centiloid approach represents a major step forward in biomarker standardization, supporting its growing adoption in multicenter trials and routine clinical evaluation of neurodegenerative diseases (6–8).

This literature review aims to evaluate the current literature on the Centiloid scale in amyloid PET imaging. By summarizing recent evidence and outlining ongoing challenges, this review seeks to highlight the Centiloid framework’s role in advancing quantitative neuroimaging and its importance in the era of precision dementia diagnostics.

## MATERIALS AND METHODS

This is a narrative review with a literature search using PubMed, Scopus, and Embase databases to identify studies published between January 2015 and October 2025, encompassing the period after the introduction of the Centiloid methodology. The following search terms were used: (“amyloid PET” OR “ $\beta$ -amyloid PET” OR “A $\beta$  PET”) AND (“Centiloid” OR “Centiloid scale” OR “Centiloid conversion”) AND (“SUVR” OR “quantification”) AND (“Alzheimer disease” OR “dementia”). Manual searches of the reference lists of included articles and relevant reviews were also performed to identify additional articles. The search was limited to peer-reviewed, English-language publications (9–42).

This study is approved by our institutional review board. The patient’s images are anonymized. We have all required licenses and permissions to use MIMneuro software, including the screenshots.

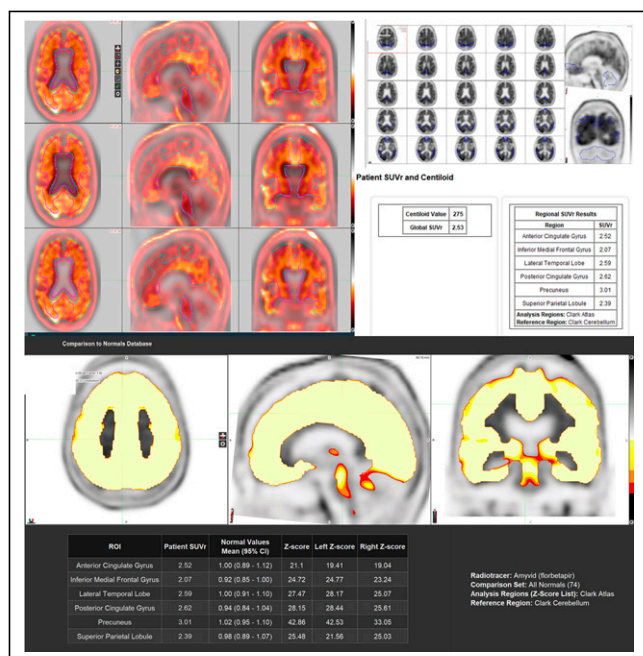
## RESULTS

### Study Characteristics

The included articles were published between 2015 and 2025, representing research across North America, Europe, Asia, and Australia. Most studies used  $^{11}\text{C}$ -PiB as the Centiloid reference tracer, whereas others focused on  $^{18}\text{F}$ -labeled tracers such as florbetapir, florbetaben, and flutemetamol. Across studies, the Centiloid methodology was consistently applied to convert SUVRs to CL units using the linear transformation derived from  $^{11}\text{C}$ -PiB-calibrated reference datasets.

### Tracers Used in Centiloid Calibration

**$^{11}\text{C}$ -PiB.**  $^{11}\text{C}$ -PiB remains the reference standard tracer for Centiloid calibration. Multiple studies validated the original Klunk et al. (1) scaling method, defining 0 CL as the mean cortical uptake in young amyloid-negative controls and 100 CL as the mean patients with mild AD. SUVRs were typically referenced to the whole cerebellum. The linear relationship between SUVR and CL was highly consistent



**FIGURE 2.** Sample report generated by MIMneuro including regional ROIs, SUVRs, and Centiloid score.

across scanners and processing pipelines, providing the foundation for cross-tracer conversions.

**<sup>18</sup>F-Florbetapir.** Florbetapir is one of the most widely studied <sup>18</sup>F tracers for Centiloid conversion. Studies by Navitsky et al. (10) demonstrated a robust linear correlation between florbetapir SUVRs and <sup>11</sup>C-PiB–derived Centiloid values ( $r^2 = 0.93$ – $0.96$ ). Reported positivity thresholds ranged between 20 and 25 CL, aligning with pathologic amyloid positivity. Florbetapir’s relatively lower dynamic range compared with that of <sup>11</sup>C-PiB was noted as a limitation but did not compromise the calibration fidelity.

**<sup>18</sup>F-Florbetaben.** Studies such as that of Rowe et al. (14) validated Centiloid scaling for florbetaben. Using paired <sup>11</sup>C-PiB and florbetaben data, a strong linear relationship was observed ( $r^2 \approx 0.95$ ). Centiloid thresholds for amyloid positivity averaged 24 CL, closely matching those derived from florbetapir studies. The consistency across cortical regions supported the robustness of Centiloid transformation, even in multisite acquisitions.

**<sup>18</sup>F-Flutemetamol.** Flutemetamol exhibited a similarly strong correlation with <sup>11</sup>C-PiB Centiloid units ( $r^2 > 0.90$ ). Studies by Battle et al. (11) derived conversion equations yielding comparable scales. The tracer demonstrated higher white matter retention, necessitating consistent preprocessing pipelines to avoid bias in SUVR calculation. Despite these challenges, the Centiloid approach effectively harmonized flutemetamol data across centers. Table 1 demonstrates approved amyloid PET tracers and key properties.

**Emerging Tracers.** A limited number of studies evaluated newer tracers (e.g., <sup>18</sup>F-NAV4694 and <sup>18</sup>F-AZD4694) (32). Early validation results show high concordance with <sup>11</sup>C-PiB–based Centiloid scales ( $r^2 \geq 0.95$ ), indicating that the Centiloid framework can be successfully extended to next-generation tracers pending formal calibration protocols.

**Reference Regions and Image Processing.** Most studies used the whole cerebellum or cerebellar cortex as the

reference region for SUVR computation, as these areas are typically devoid of amyloid deposition. Alternative references, including the pons or composite subcortical regions, were used in select studies to improve stability in longitudinal analyses or hybrid PET/MR acquisitions. Differences in preprocessing pipelines (SPM vs. PMOD vs. FreeSurfer) introduced minor variability, typically within  $\pm 3$  CL, confirming that the Centiloid framework remains robust across analysis platforms if standard calibration parameters are applied. Figure 3 demonstrates a 3-dimensional stereotactic surface projection image processed by MIMneuro.

**Centiloid Thresholds for Amyloid Positivity.** Across all included studies, Centiloid thresholds defining amyloid positivity ranged from 16 to 30 CL, with the most frequently cited cutoff between 20 and 25 CL. These thresholds corresponded to approximately 20 CL for early pathologic amyloid accumulation, greater than 50 CL for cognitive impairment or clinical AD range, and greater than 80 CL for advanced plaque burden.

Several studies correlated Centiloid values with cerebrospinal fluid A $\beta$ 42 concentrations, tau PET uptake, and cognitive scores (e.g., Mini-Mental State Examination or Clinical Dementia Rating), demonstrating that a threshold near 20 CL aligns with the onset of abnormal amyloid biochemistry. This finding supports the Centiloid scale as a clinically meaningful biomarker of disease stage and therapeutic response. Table 2 demonstrates Centiloid threshold ranges across studies.

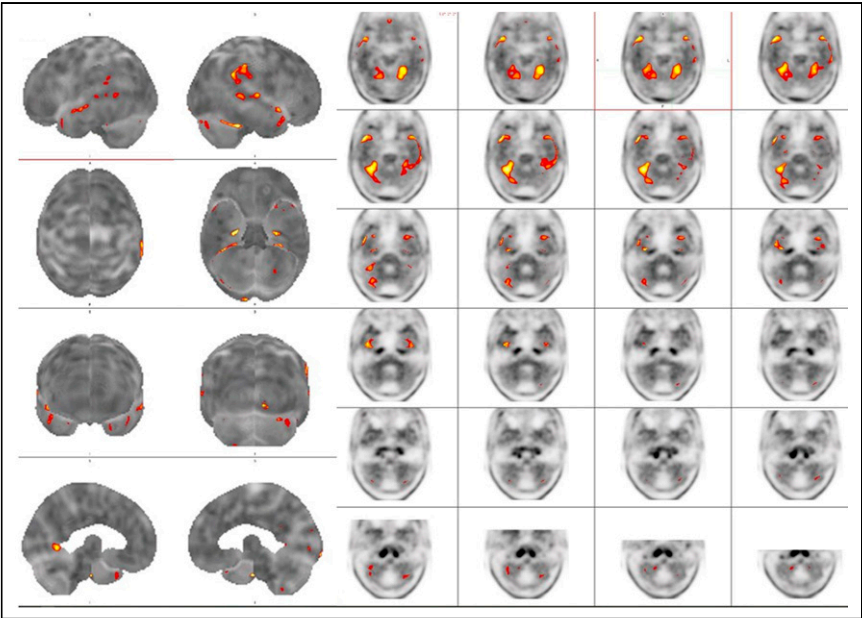
**Clinical and Research Application.** Implementation of the Centiloid scale has enabled multicenter data pooling, allowing direct comparison across tracers and sites in large consortia such as Alzheimer’s Disease Neuroimaging Initiative, Australian Imaging, Biomarker and Lifestyle Study on Aging, and European Prevention of Alzheimer’s Dementia; longitudinal monitoring, facilitating quantitative assessment of amyloid accumulation over time and response to anti-amyloid therapies (e.g., aducanumab, lecanemab); integration into the AT(N) framework, providing the “A” (amyloid) component for composite biomarker models alongside tau and

**TABLE 1**  
Approved Amyloid PET Tracers and Key Properties

| Tracer                                                         | Isotope         | Half-life (min) | Dynamic range | White matter retention | Reference region              | Recommended acquisition window (min) |
|----------------------------------------------------------------|-----------------|-----------------|---------------|------------------------|-------------------------------|--------------------------------------|
| <sup>11</sup> C-PiB                                            | <sup>11</sup> C | 20              | High*         | Low                    | Cerebellar cortex             | 40–60                                |
| <sup>18</sup> F-florbetapir (Amyvid; Lilly)                    | <sup>18</sup> F | 110             | Moderate      | High                   | Whole cerebellum/white matter | 30–50                                |
| <sup>18</sup> F-florbetaben (Neuraceq; Life Molecular Imaging) | <sup>18</sup> F | 110             | Moderate–high | Moderate               | Cerebellar gray matter        | 90–110                               |
| <sup>18</sup> F-flutemetamol (Vizamyl; GE HealthCare)          | <sup>18</sup> F | 110             | Moderate      | High                   | Whole cerebellum              | 90–110                               |
| <sup>18</sup> F-NAV4694                                        | <sup>18</sup> F | 110             | Very high†    | Low                    | Cerebellar cortex             | 50–70                                |

\*Excellent cortex–background contrast, rapid kinetics.

†<sup>11</sup>C-PiB–like behavior.



**FIGURE 3.** Three-dimensional stereotactic surface projection image process by MIMneuro.

neurodegeneration imaging; regulatory and clinical adoption, with Centiloid-referenced reporting increasingly recommended for quantitative PET in dementia evaluation.

Overall, evidence demonstrates that the Centiloid framework substantially enhances reproducibility and interpretability of amyloid PET data across imaging centers, tracers, and clinical trials. Figure 4 demonstrates pre- and posttreatment

brain PET images with Centiloid values after receiving an anti-amyloid agent, with significant change in amyloid burden and a decrease in Centiloid value.

**Limitations.** Although widely adopted, several studies noted sources of variability, including dependence on accurate reference region segmentation; white matter spillover effects, especially in  $^{18}\text{F}$  tracers; limited standardization for hybrid PET/MR systems; and the need for calibration updates with newer tracers and reconstruction algorithms.

Despite these, the Centiloid system consistently improved data harmonization compared with unscaled SUVR measurements.

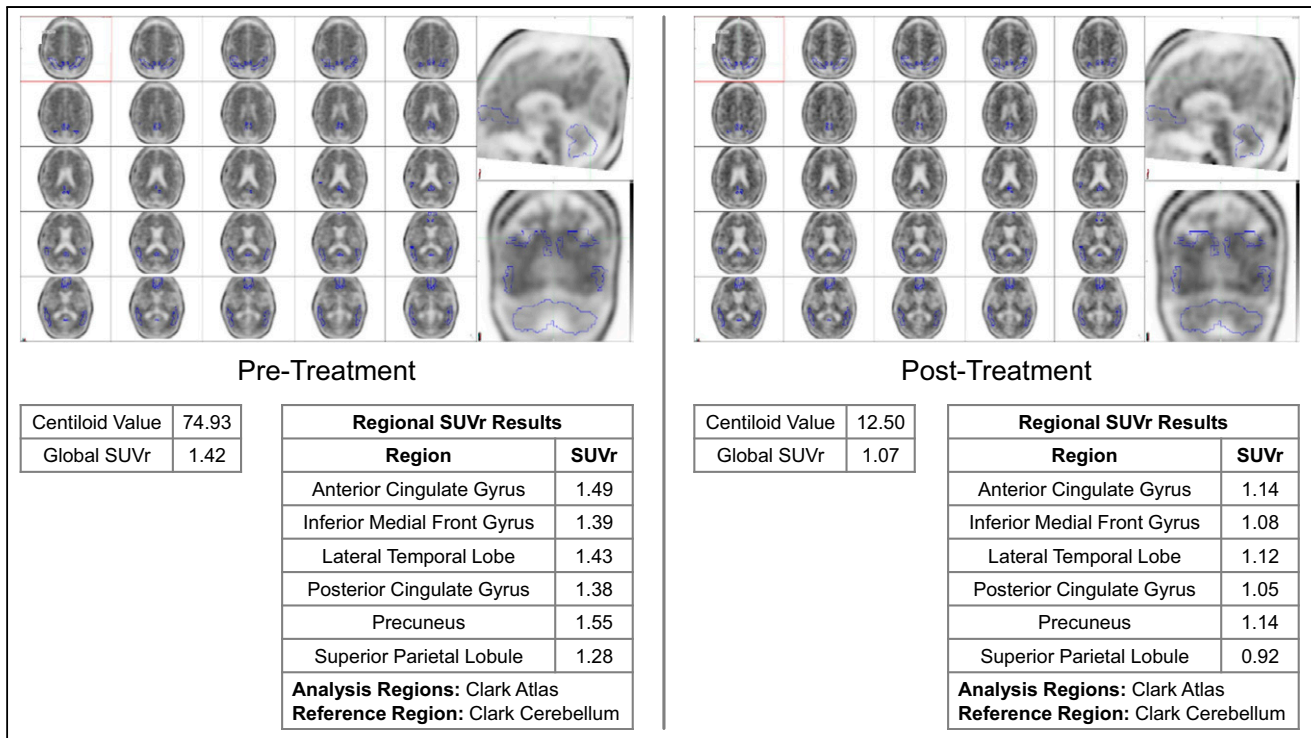
### DISCUSSION

This article reviews studies evaluating the Centiloid scale as a standardized framework for quantifying A $\beta$  burden in PET imaging for AD. Since its introduction in 2015 by Klunk et al., the Centiloid methodology has gained broad acceptance as a universal metric for cross-tracer and cross-site harmonization. This review confirms that the Centiloid scale effectively addresses one of the most significant limitations in amyloid imaging: the

**TABLE 2**  
Centiloid Threshold Ranges Across Major Amyloid PET Cohorts

| Cohort or study    | Centiloid threshold (CL) | Interpretation and use                                            | Tracer context                                 | Key reference |
|--------------------|--------------------------|-------------------------------------------------------------------|------------------------------------------------|---------------|
| ADNI               | <10                      | Amyloid-negative (no significant neuritic plaques)                | $^{11}\text{C}$ -PiB, florbetapir, florbetaben | (43)          |
|                    | 15–18                    | Earliest amyloid accumulation; predicts future A $\beta$ increase | PiB $\rightarrow$ CL                           | (44)          |
|                    | 20–25                    | Amyloid positivity (commonly used cutoff)                         | Harmonized CL                                  | (1)           |
| AIBL               | >30                      | Robust amyloid burden                                             | —                                              | (43)          |
|                    | <10                      | Amyloid-negative                                                  | PiB, $^{18}\text{F}$ tracers                   | (45)          |
|                    | 15–18.5                  | Low-level/early amyloid; prognostic threshold                     | PiB $\rightarrow$ CL                           | (44)          |
|                    | ~25                      | Standard amyloid positivity                                       | CL-scaled                                      | (9,14)        |
| OASIS              | $\geq 35$ –40            | Established amyloid pathology                                     | —                                              | (9,14)        |
|                    | <10                      | Amyloid-negative                                                  | Florbetapir, CL-scaled                         | (47)          |
|                    | 15–20                    | Intermediate/gray zone                                            | —                                              | (48)          |
|                    | 24–25                    | Amyloid-positive (aligned with ADNI/AIBL)                         | —                                              | (43)          |
| AMYPAD (DPMS/PNHS) | >30                      | High amyloid burden                                               | —                                              | (1)           |
|                    | <10                      | Amyloid-negative (pathology-negative)                             | Visual + quantitative                          | (7,49)        |
|                    | ~12                      | Earliest detectable amyloid pathology                             | Multitracer CL                                 | (7,49)        |
|                    | 21–24                    | Intermediate amyloid/reader-challenging cases                     | —                                              | (7,49)        |
|                    | ~25                      | Typical amyloid positivity                                        | —                                              | (7,49)        |
|                    | 30–40                    | Clear amyloid positivity (clinical confidence)                    | —                                              | (7,49)        |

ADNI = Alzheimer’s Disease Neuroimaging Initiative; AIBL = Australian Imaging, Biomarkers and Lifestyle Study on Aging; OASIS = Open Access Series of Imaging Studies; AMYPAD = Amyloid Imaging to Prevent Alzheimer’s Disease; DPMS = Diagnostic and Patient Management Study; PNHS = Prognostic and Natural History Study.



**FIGURE 4.** Pre- and posttreatment brain PET images with Centiloid values after receiving anti-amyloid agent with significant change in amyloid burden and decrease in Centiloid value.

lack of comparability among tracers, scanners, and analysis pipelines, thereby enhancing the reproducibility and interpretability of amyloid PET data across clinical and research settings.

### Standardization and Reproducibility

Traditional measures of amyloid burden, such as SUVRs, have been prone to variability, resulting from differences in scanner characteristics, image reconstruction algorithms, and reference region selection. The Centiloid scale offers a straightforward yet powerful linear transformation that expresses all amyloid PET data on a common metric, with 0 CL representing the mean value in young amyloid-negative controls and 100 CL corresponding to the mean in patients with mild AD. This standardization allows data derived from diverse tracers including  $^{11}\text{C}$ -PiB,  $^{18}\text{F}$ -florbetapir,  $^{18}\text{F}$ -florbetaben, and  $^{18}\text{F}$ -flutemetamol to be directly compared on a unified scale.

Across the studies reviewed, Centiloid conversion yielded strong linear correlations ( $r^2 > 0.9$ ) between SUVRs from  $^{18}\text{F}$ -labeled tracers and the  $^{11}\text{C}$ -PiB reference, with intersite coefficients of variation typically below 5%. These findings indicate that Centiloid scaling not only harmonizes quantitative data but also preserves biologic sensitivity to amyloid pathology. Importantly, the Centiloid framework remains robust across variations in image processing platforms (SPM, PMOD, FreeSurfer) if calibration datasets and reference regions are standardized.

### Clinical and Research Applications

The Centiloid scale has emerged as an invaluable tool in both clinical practice and multicenter research. In clinical contexts, quantitative Centiloid values supplement visual interpretation, enhancing diagnostic confidence, particularly in borderline or early-stage cases. A threshold near 20–25 CL has consistently been identified as the point at which cortical amyloid accumulation exceeds normal aging levels, correlating with reductions in cerebrospinal fluid A $\beta$ 42 and with early cognitive decline. These thresholds have been incorporated into diagnostic algorithms and clinical trials targeting early AD intervention.

In research, the Centiloid scale has become essential for multicenter studies and longitudinal designs, such as the Alzheimer's Disease Neuroimaging Initiative, Australian Imaging, Biomarker and Lifestyle Study on Aging, and European Prevention of Alzheimer's Dementia. By providing a consistent quantitative reference, Centiloid conversion enables pooling of amyloid PET data across tracers, enhancing statistical power and reproducibility. Moreover, the use of Centiloid units facilitates longitudinal tracking of amyloid accumulation over time, allowing estimation of annualized change rates in Centiloids (typically 1–2 CL/y in preclinical cohorts) (2). This standardization has proven invaluable for monitoring therapeutic response in anti-amyloid clinical trials, including those evaluating aducanumab, donanemab, and lecanemab, where Centiloid reduction serves as an objective biomarker of target engagement.

Centiloid thresholds for amyloid PET positivity across major biomarker cohorts and approximate cutoffs are based on published standardization and validation efforts: Australian Imaging, Biomarker and Lifestyle Study on Aging (~20 CL), Alzheimer's Disease Neuroimaging Initiative (~22 CL), Open Access Series of Imaging Studies (~25 CL), and Amyloid Imaging to Prevent Alzheimer's Disease (~22 CL). Overall, a Centiloid value of more than 30 may be considered a reliable cutoff for enrollment in trials investigating anti-amyloid monoclonal antibodies such as lecanemab. Minor variations reflect cohort-specific harmonization and assay differences.

The Centiloid system has also been integrated into the AT(N) biomarker framework, serving as the "A" (amyloid) component alongside tau PET and neurodegeneration markers such as FDG PET or structural MRI. This triad enables more precise stratification of disease stage and supports the transition toward biologically defined definitions of AD. Thus, the adoption of Centiloid quantification represents a critical advance in the standardization of biomarker-driven diagnosis and treatment evaluation.

### Methodologic Considerations and Limitations

Although the Centiloid framework has demonstrated strong intertracer consistency, several methodologic challenges persist. One key consideration is the selection of reference regions, as different reference choices (whole cerebellum, cerebellar cortex, or pons) can subtly affect SUVR and, consequently, Centiloid scaling. Most studies recommend maintaining the whole cerebellum as the standard reference to ensure compatibility with the original  $^{11}\text{C}$ -PiB-based calibration.

Another limitation involves white matter retention, particularly with  $^{18}\text{F}$ -labeled tracers such as florbetapir and flutemetamol, which may introduce variability due to nonspecific binding. Consistent image preprocessing, including partial-volume correction and white matter masking, is therefore critical to maintain reliable Centiloid estimates. Additionally, scanner harmonization remains an area of active investigation, as differences in spatial resolution and reconstruction algorithms can influence SUVR calculations before Centiloid conversion. Ongoing initiatives such as the European Association of Nuclear Medicine Research Ltd. accreditation and the Centiloid Project calibration database continue to address these standardization needs.

Specific attention should be exercised when drawing and checking ROIs. Inclusion of bone or cerebrospinal fluid will decrease the mean SUV in an ROI, while inclusion of white matter will increase it. Also, inclusion of brainstem or occipital lobe in the cerebellar ROI will increase the mean SUV. Any change in the measured SUV can be translated to change in Centiloid value as they have positive and linear correlation.

The anatomic brain images (CT and MRI) should be evaluated for any anatomic or physiologic defect. Atrophy, stroke, or other defects will decrease the mean SUV in an ROI.

The effect of higher mean SUV in an analysis region is increased global SUVR and possible false positive and vice versa. The effect of higher mean SUV in the cerebellar reference region is a lower global SUVR and a possible false negative and vice versa.

Any high physiologic uptake outside brain, for example, in salivary glands, may result in misplaced ROIs by the software and thus incorrect Centiloid values. Additional technical factors which may affect image interpretation include but are not limited to change in radiotracer dose and uptake time, image noise due to motion artifact, partial-volume effect leading to low gray/white matter contrast, and low signal-to-noise ratio. Furthermore, in the borderline cases, reporting regional SUVR and  $z$  scores as well as Centiloid may be considered.

Finally, whereas the Centiloid scale currently encompasses the major amyloid tracers, emerging agents (e.g.,  $^{18}\text{F}$ -NAV4694 and  $^{18}\text{F}$ -MK-6240) require additional cross-validation studies. Similarly, as hybrid PET/MR systems and deep-learning-based quantification tools become more prevalent, recalibration and methodologic updates will be essential to preserve cross-platform comparability.

### Comparison with Other Quantification Frameworks

Alternative standardization methods, such as  $z$ -scoring approaches, global SUVR normalization, and image-based harmonization, have been proposed but lack the broad validation and simplicity of the Centiloid system. The Centiloid framework's advantage lies in its universality and interpretability. A Centiloid value of 0 CL represents no amyloid, and 100 CL represents typical AD levels, providing a biologically intuitive and clinically communicable metric. Unlike arbitrary SUVR scales, Centiloid units allow straightforward cross-study comparison and threshold interpretation, making them increasingly favored in both academic and regulatory contexts.

### CONCLUSION

The Centiloid scale has become a cornerstone of standardized amyloid PET quantification, transforming the way A $\beta$  burden is measured, compared, and interpreted in AD research and clinical practice. By providing a common quantitative language across tracers, scanners, and analytic methods, the Centiloid framework enables robust multicenter data integration, facilitates biomarker-driven clinical trials, and enhances diagnostic confidence in dementia imaging.

Although methodologic refinements remain necessary, particularly for emerging tracers and hybrid imaging platforms, the evidence consistently supports the reliability and clinical relevance of Centiloid quantification. As neuroimaging moves toward precision medicine, the Centiloid system stands as a model of successful standardization, bridging the gap between research innovation and clinical application in the fight against AD.

## DISCLOSURE

No potential conflict of interest relevant to this article was reported.

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